BARANGAY BANLIC CCIR SYSTEM

Citizen Complaint & Incident Reporting System | City of Cabuyao, Laguna

TECHNICAL ARCHITECTURE & CAPSTONE DEFENSE GUIDE: DATA ANALYTICS

1. EXECUTIVE OVERVIEW & STRATEGIC PURPOSE

In the **Barangay Banlic CCIRS**, **Data Analytics** transforms isolated, transactional complaints into actionable governance intelligence. Rather than treating resident feedback merely as ad-hoc tickets, the system aggregates operational data to empower barangay executives and committee chairpersons to transition from *reactive problem solving* to *proactive, evidence-based governance*.

Real-Time	6 Puroks	>48h SLA	100% Export
FILING VELOCITY	GEOGRAPHIC HOTSPOTS	TRIAGE SAFEGUARD	EXCEL & BI READY

2. ACADEMIC & OPERATIONAL ANALYTICS FRAMEWORK

The system implements the standard three-tier data analytics framework:

Analytics Tier	Core Question Addressed	Implementation in Banlic-CCIRS
1. Descriptive Analytics	*What is happening in the barangay right now?*	Real-time counters for submissions, in-progress cases, and resolution rates. Categorical distribution across noise, sanitation, and peace & order.
2. Diagnostic Analytics	*Where are the delays and localized incident concentrations?*	Geospatial Purok Bar Charts identifying community hotspots. Automated >48h Overdue SLA indicators flagging triage bottlenecks.
3. Prescriptive / Trend Insights	*What preventive actions should the council take?*	6-Month historical rolling trendlines detecting seasonal spikes, guiding proactive Tanod patrols and localized municipal budget allocations.

> **Key Operational Value:** When sanitation or road obstruction complaints surge in a specific sitio (e.g. Purok 4), the barangay council uses the analytics engine to immediately deploy dedicated waste collection rounds and tanod monitoring, resolving root causes before issues escalate.

3. DATA VISUALIZATION ARCHITECTURE

The visualization suite is built on **Chart.js** with bidirectional reactive theme synchronization (supporting both Light and Dark modes):

- **6-Month Report Volume Trend (Smooth Line Chart):** Plots monthly submission trajectory across a rolling window to detect macro patterns.
- **Resolution Status Lifecycle (Doughnut Chart):** Color-coded breakdown across all stages (Pending Review, Under Review, Assigned, In Progress, Resolved, Closed, Rejected).
- **Geographical Density by Purok / Area (Bar Chart):** Quantitative comparison of incident frequency per sitio to guide on-the-ground patrol schedules.
- **Categorical Distribution (Bar Chart):** Categorizes incidents by administrative committee (Peace & Order, Infrastructure, Sanitation, Women & Children).

4. MULTI-DIMENSIONAL QUERYING & DATA INTEROPERABILITY

To support rigorous auditing, academic research, and municipal reporting, the analytics module includes:

- **Dynamic Multi-Factor Query Engine:** Enables administrators to filter records concurrently by date range (`date_from`, `date_to`), category, status, priority, and purok with immediate dynamic metric re-calculation.
- **Automated CSV Export:** Features UTF-8 BOM encoding for direct, zero-configuration compatibility with *Microsoft Excel*, *Google Sheets*, *PowerBI*, and *Tableau*.
- **Executive Print & PDF Summary:** Generates formal printable audit sheets suitable for monthly Barangay Council sessions and DILG compliance reports.

5. CAPSTONE DEFENSE & PRESENTATION SCRIPT

Sample 60-Second Presentation Delivery:
"Good day, members of the panel. In our Barangay Banlic CCIRS, Data Analytics is the core driver of modern local governance. Instead of handling complaints in isolation, our analytics engine aggregates submissions across all puroks and committees.

Through interactive Chart.js visualizations, barangay officials monitor monthly volume trends, measure response velocity, and immediately spot incident hotspots. If data indicates an influx of public safety reports in Purok 2, the administration can proactively assign tanod patrols rather than waiting for formal hearings. The module also enforces a 48-hour SLA triage alert and supports full CSV data export for municipal reporting."

6. ANTICIPATED PANEL QUESTIONS & ANSWERS (Q&A DEFENSE GUIDE)

Q1: How does this qualify as Data Analytics rather than just a database report?

A: Simple reporting merely lists past events. Banlic-CCIRS applies analytical computation—including multi-dimensional cross-tabulations, rolling 6-month trend lines, geographic hotspot detection, and automated threshold alerts (>48h SLA warnings) that diagnose bottlenecks and prescribe targeted operational deployments.

Q2: How does the analytics engine protect citizen privacy under the Data Privacy Act?

A: All public-facing analytics and reference tracking utilize anonymized aggregations. Complainant names are cryptographically masked (e.g., J*** D***), and sensitive evidentiary files are stored in secured local disks accessible strictly to authenticated case handlers.

Q3: Can this data be utilized by external local government agencies like the Cabuyao City LGU?

A: Yes. The system features a native CSV streaming engine with UTF-8 encoding that outputs complete structured datasets, allowing seamless ingestion into municipal GIS, DILG reporting frameworks, and external business intelligence tools.